Chapter 15: The Deflation Bias

Gauti B. Eggertsson, Brown University, June 2, 2020

1 Overview

2 The Deflation Bias

This section builds on Eggertsson (2006). We abstract from the inflation bias, i.e. inflation bias $x^* = 0$.

We do not take into account that the zero bound might be binding. In this case, then the IS curve does not simply back out the interest rate needed to implement a particular equilibrium. Instead, it plays an important role by determining aggregate demand.

The model, as before, is given by:

$$U_t \approx E_t \sum_{t=0}^{\infty} \beta^t \{\pi_t^2 + \lambda x_t^2\} \tag{1}$$

$$\pi_t = \kappa x_t + \beta E_t \pi_{t+1} \tag{2}$$

$$x_t = E_t x_{t+1} - \sigma(\hat{\imath}_t - E_t \pi_{t+1} - \hat{r}_t^e) \tag{3}$$

$$\hat{\imath}_t + \beta \bar{\imath} \geq 0 \tag{4}$$

Here the $\bar{\imath}$ corresponds to the the nominal interest rate in the zero inflation steady state that we approximate around. The zero bound takes on this form since $\hat{\imath}_t$ is expressed in deviation from the zero inflation steady state.

The optimal commitment solution can be characterized via the following Lagrangian:

$$\begin{aligned} L_t &= E_t \sum_{s=t}^{\infty} \beta^{s-t} \{\frac{1}{2}\pi_s^2 + \frac{\lambda}{2} {x_s}^2 + \phi_{1s}\{x_s - x_{s+1} + \sigma(i_s - \pi_{s+1} - r_s^n\} \\ &\quad + \phi_{2s}\{\pi_s - \kappa x_s - \beta \pi_{s+1}\} + \phi_{3t}(\hat{\imath}_t + \beta \bar{\imath}) \end{aligned}$$

The first-order conditions are:

$$\partial\pi \qquad \pi_t - \beta^{-1}\sigma\phi_{1t-1} + \phi_{2t} - \beta^{-1}\phi_{2t-1} = 0$$

$$\partial x \qquad \lambda x_t + \phi_{1t} - \beta^{-1}\phi_{1t-1} - \kappa\phi_{2t} = 0$$

$$\partial i \qquad \phi_{1t} + \phi_{3t} = 0$$

$$\phi_{3t} \geq 0, \quad \hat{\imath}_t + \beta\bar{\imath} \geq 0, \quad (\hat{\imath}_t + \beta\bar{\imath})\phi_{3t} = 0$$

The last inequalities are Kuhn-Tucker complementary slackness conditions.

The optimal discretion solution can be characterized via the following Lagrangian:

$$\begin{aligned} L_t \quad = \quad & E_t \sum_{s=t}^{\infty} \beta^{s-t}\{\frac{1}{2}\pi_s^2 + \frac{\lambda}{2}{x_s}^2 + \phi_{1s}\{x_s - \bar{x} + \sigma(i_s - \bar{\pi} - r_s^n\} \\ & + \phi_{2s}\{\pi_s - \kappa x_s - \beta\bar{\pi}\} + \phi_{3t}(\hat{\imath}_t + \beta\bar{\imath}) \end{aligned}$$

$$\partial\pi \qquad \pi_t + \phi_{2t} = 0$$

$$\partial x \qquad \lambda x_t + \phi_{1t} - \kappa\phi_{2t} = 0$$

$$\partial i \qquad \phi_{1t} + \phi_{3t} = 0$$

$$\phi_{3t} \geq 0, \quad (\hat{\imath}_t + \beta\bar{\imath}) \geq 0, \ (\hat{\imath}_t + \beta\bar{\imath})\phi_{3t} = 0$$

The thought experiment is as follows: The efficient rate of interest drops in period zero to some low level $\hat{r}_t^e = \hat{r}_S^e$ (S denotes "short-run") where we have in mind the stories modeled in the last section. In every period $t > 0$ there is a fixed probability $1 - \mu$ that the shock reverts back to its steady state, $\hat{r}_L^e = 0$. Thus at time 0 there is a $1-\mu$ chance the shock is "over" in period 1, while if the the shock remains in period 1 the probability that the shock is over in period 2 is once again $1 - \mu$. We call the stochastic date at which the normalization occurs t_L and assume that once back to normal the shock stays there. It is easy to confirm, then, that the expected duration of the shock/recession is $1/(1-\mu)$.

Solution: The next figure shows with a solid line commitment and with dashed line the MPE. It shows only one "contingency", i.e. the case in which the shock reverts back to normal in period 15.

The discretion solution is dashed and labeled $\pi^* = 0$ The solid solution is the commitment.

As you can see, the discretion leads to massive drops in output and inflation. Meanwhile the commitment implies much less drop. The commitment does this by committing to keeping interest rate low for a considerable time. This lessens the output drop and the deflation.

The problem, however, is that this policy is not dynamically consistent. To see why, notice that from the time period the shock is over onwards (period $t > 15$), then the discretionary government hits the bliss-point of zero output gap and zero inflation, while the government operating under commitment has excessive inflation and and output boom. This creates a credibility problem.

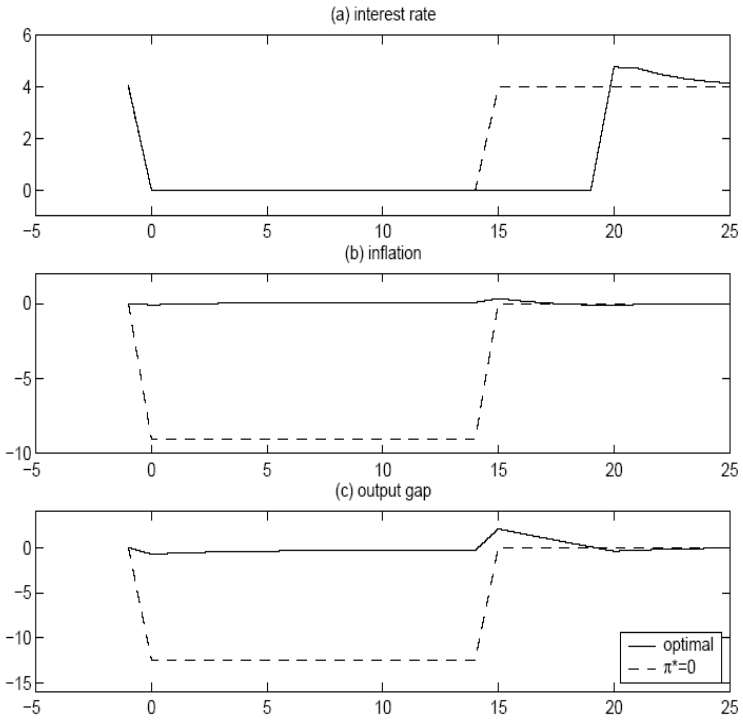


Figure 1: The Deflation Bias